AI Playbook and Toolkit for Public Health Departments

Data Pipelines for Public Health AI

ARC 210

Data pipelines are the operational pathways that move public health data from source systems into analytic, reporting, and workflow environments. AI tools depend on these pathways even when users do not see them. A model that classifies cases, summarizes laboratory reports, or predicts service demand can only work responsibly if the data feeding the model are timely, complete, transformed correctly, and monitored. This module helps learners understand the pipeline steps that shape AI outputs and the safeguards needed when pipeline failures could affect public health action.

Level	applied/foundational
Primary track	Technical Architecture Track
Appears in tracks	technical-architecture

Learning Objectives

- Describe the major steps in public health data pipelines, including ingestion, validation, transformation, storage, access, and monitoring.
- Distinguish between batch, real-time, and event-driven data flows.
- Explain how pipeline failures can affect AI outputs and operational decisions.
- Identify metadata, lineage, version control, and audit needs for AI-supported workflows.
- Produce a data pipeline readiness checklist for a public health AI use case.

Module Overview

Data pipelines are the operational pathways that move public health data from source systems into analytic, reporting, and workflow environments. AI tools depend on these pathways even when users do not see them. A model that classifies cases, summarizes laboratory reports, or predicts service demand can only work responsibly if the data feeding the model are timely, complete, transformed correctly, and monitored. This module helps learners understand the pipeline steps that shape AI outputs and the safeguards needed when pipeline failures could affect public health action.

Why This Topic Matters for Public Health AI

Data pipeline readiness matters because AI systems can amplify routine data management problems. A delayed file, changed field name, failed interface, missing date, duplicate record, or outdated transformation rule may not be obvious to an end user. The AI tool may continue generating fluent summaries or confident classifications even after the data feed has become unreliable.

Public health agencies need pipeline visibility before they rely on AI for operational workflows. Visibility means knowing where data originate, how often they update, what transformations occur, what validation checks run, and who is alerted when something fails. A pipeline should not be treated as a hidden technical detail. It is part of the safety infrastructure for AI.

Pipeline review also supports public accountability. When an AI output affects case prioritization, outreach, resource allocation, or public messaging, the agency may need to explain which data were used and whether those data were current. A well-documented pipeline makes that explanation possible.

Core Concepts

Data pipeline. A data pipeline is a set of technical and operational steps that moves data from a source to a destination where the data can be used. In public health, pipelines may include laboratory interfaces, EHR feeds, IIS submissions, syndromic surveillance feeds, vital records extracts, HIE connections, and program datasets. Each step creates an opportunity to improve or degrade data quality.

ETL and ELT. ETL stands for extract, transform, load, while ELT stands for extract, load, transform. ETL transforms data before they enter the target system, while ELT loads data first and transforms them later. The distinction matters for AI because staff need to know where data were changed and which version was used by the AI tool.

Pipeline monitoring. Pipeline monitoring is the routine process of checking whether data feeds, transformations, and outputs are working as intended. Monitoring may include file receipt, schema validation, required-field checks, timeliness checks, duplicate review, error queues, and volume trends. Monitoring should be tied to escalation when failures affect AI use.

Data lineage. Data lineage shows where data came from, what happened to them, and where they were used. Lineage supports validation, troubleshooting, incident response, and public records review. In AI workflows, lineage helps staff trace a questionable output back to its source data and transformation history.

Public Health Example

A syndromic surveillance team may use AI to identify unusual emergency department visit patterns. The model depends on timely data feeds from hospitals, correct parsing of chief complaint text, reliable facility identifiers, and consistent geographic information. If one large hospital feed stops or a field changes after an EHR upgrade, the AI tool may miss a signal or generate a false alert. Pipeline monitoring is therefore part of outbreak detection readiness.

A second example is an AI tool that drafts summaries of lab-based cases. The tool may rely on an ELR feed, a terminology mapping table, a deduplication process, and a case management interface. If any of those steps fail, the summary may omit a result, duplicate a patient, or misrepresent reportability. The pipeline must be reviewed before the AI output can be trusted.

Technical Deep Dive

Pipeline design begins with the source and ends with the user-facing decision or output. In between, data may be received, staged, validated, transformed, standardized, linked, loaded, indexed, analyzed, and displayed. Each step should have an owner, a purpose, a quality check, and a documented failure process.

Batch pipelines move data on a schedule, such as nightly files or weekly extracts. Real-time pipelines move data as events occur, such as laboratory reports or eCR messages. Event-driven pipelines trigger actions when a defined event occurs, such as creating a review task when a high-priority result arrives. AI workflows may use any of these models, but the monitoring and risk profile differ for each.

Metadata and version control are essential for AI-supported pipelines. Staff need to know which data dictionary, code mapping, model version, prompt template, transformation rule, and document collection were used at a specific time. This is especially important when an agency investigates an incident or compares AI performance before and after a system change.

Risks, Failure Modes, and Guardrails

Silent feed failure. A data feed can stop or change without visible failure in the AI interface. Guardrails include automated feed monitoring, expected volume thresholds, error queues, and notification procedures.

Schema and terminology changes. Source systems may change field names, formats, codes, or units. Guardrails include schema validation, terminology governance, release notes, and regression testing.

Unclear fallback process. Staff may continue using AI outputs when data are stale or incomplete. Guardrails include visible freshness indicators, pause rules, and manual fallback procedures.

Weak lineage. If transformations are not documented, staff may not be able to explain or correct AI errors. Guardrails include lineage documentation, version control, and audit trails.

Application to AI-Supported Workflows

Generative AI may use pipeline outputs to summarize records, but the summary should indicate source freshness and preserve traceability. Predictive models may rely on engineered features created by a pipeline, so the feature pipeline must be monitored for drift and missingness. RAG systems depend on document ingestion pipelines, which must track document source, version, permissions, and refresh schedules. Agentic workflows may depend on event-driven pipelines, making failure handling and approval points especially important.

Reflection Questions

Which public health data feeds are most important to daily decisions in your program?

Which pipelines are automatically monitored, and which depend on manual review?

What failures would materially affect an AI output?

How would staff know if an AI output was based on stale or incomplete data?

Who has authority to pause AI use when a pipeline fails?

Practical Exercise

Create a data pipeline readiness checklist for one AI-supported workflow. Identify the data source, update frequency, required fields, validation checks, transformations, lineage documentation, monitoring indicators, error notification process, and fallback procedure.

Document at least three pipeline failure scenarios and describe how each should be detected, escalated, and communicated to users.

Expected Artifact or Evidence

Data pipeline map

Pipeline readiness checklist

Failure and escalation plan

Known data limitations statement

Expected Assignment

- Data pipeline map
- Pipeline readiness checklist
- Failure and escalation plan
- Known data limitations statement

Knowledge Check

- 1. What is the strongest reason to document data lineage for an AI-supported workflow?
- 2. Which pipeline type is most likely to trigger a workflow immediately after a defined event?
- 3. What is a practical guardrail for silent feed failure?
- 4. Why should source freshness be visible to users?
- 5. What is strong evidence of completion for this module?

References and Resources

- CDC Data Modernization Initiative materials: https://www.cdc.gov/data-modernization/php/index.html
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- Agency data governance and system monitoring documentation